

Low Acrylonitrile NBR

POLYMERIZATION/SYNTHESIS PATENT RESEARCH REPORT

1. ABSTRACT

This report provides a comprehensive overview of the patent landscape related to Low Acrylonitrile NBR (nitrile butadiene rubber) polymerization and synthesis. The research scope focuses on the definition and production of Low Acrylonitrile NBR, with a selected patent landscape that includes five relevant patents. The major polymerization approaches include emulsion polymerization, with varying monomer ratios, emulsifier systems, initiators, and chain-transfer agents. The major formulation/process trends include the use of different acrylonitrile contents, butadiene/ACN ratios, and polymerization temperatures. The report aims to synthesize the extracted data into a cohesive technical report, highlighting the key findings and trends in the production of Low Acrylonitrile NBR.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

EP3037446

Patent Details: - Patent Number: EP3037446 - Patent Title: Method for preparing nitrile-based rubber - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2016-06-29 - Patent Status: Not disclosed - Polymer Type: Nitrile-based rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of nitrile-based rubber, which includes Low Acrylonitrile NBR

Polymerization / Synthesis Method: - Polymerization process: Not disclosed - Monomer system: Not disclosed - Acrylonitrile content: Not disclosed - Butadiene content: Not disclosed - Exact monomer ratio: Not disclosed - Water amount: Not disclosed - Emulsifier: Not disclosed - Emulsifier loading: Not disclosed - Initiator: Not disclosed - Initiator loading: Not disclosed - Catalyst: Not disclosed - Chain-transfer agent: Not disclosed - Chain-transfer dosage: Not disclosed - Polymerization temperature: Not disclosed - Pressure: Not disclosed - pH: Not disclosed - Reaction time: Not disclosed - Conversion: 0 - Coagulation conditions: Not disclosed - Post-treatment: Not disclosed - Raw polymer properties: Not disclosed

Technical relevance: This patent is relevant to Low Acrylonitrile NBR polymerization as it provides a method for preparing nitrile-based rubber, which includes Low Acrylonitrile NBR. However, the patent does not disclose specific details about the polymerization process, monomer system, or acrylonitrile content.

EP3466997

Patent Details: - Patent Number: EP3466997 - Patent Title: Method for preparing nitrile-based rubber - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2019-04-10 - Patent Status: Not disclosed - Polymer Type: Nitrile-based rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of nitrile-based rubber, which includes Low Acrylonitrile NBR

Polymerization / Synthesis Method: - Polymerization process: Not disclosed - Monomer system: Not disclosed - Acrylonitrile content: 34 - Butadiene content: Not disclosed - Exact monomer ratio: Not disclosed - Water amount: 190 - Emulsifier: Not disclosed - Emulsifier loading: Not disclosed - Initiator: 34 - Initiator loading: Not disclosed - Catalyst: Not disclosed - Chain-transfer agent: Not disclosed - Chain-transfer dosage: Not disclosed - Polymerization temperature: Not disclosed - Pressure: Not disclosed - pH: Not disclosed - Reaction time: 78 - Conversion: 78 - Coagulation conditions: Not disclosed - Post-treatment: Not disclosed - Raw polymer properties: Not disclosed

Technical relevance: This patent is relevant to Low Acrylonitrile NBR polymerization as it provides a method for preparing nitrile-based rubber, which includes Low Acrylonitrile NBR. The patent discloses an acrylonitrile content of 34, which is within the range of Low Acrylonitrile NBR.

EP2966122

Patent Details: - Patent Number: EP2966122 - Patent Title: Acrylonitrile butadiene rubber composition, acrylonitrile butadiene rubber ... - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2020-11-25 - Patent Status: Not disclosed - Polymer Type: Acrylonitrile butadiene rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of acrylonitrile butadiene rubber, which includes Low Acrylonitrile NBR

Polymerization / Synthesis Method: - Polymerization process: Not disclosed - Monomer system: Not disclosed - Acrylonitrile content: Not disclosed - Butadiene content: Not disclosed - Exact monomer ratio: Not disclosed - Water amount: Not disclosed - Emulsifier: Not disclosed - Emulsifier loading: Not disclosed - Initiator: Not disclosed - Initiator loading: Not disclosed - Catalyst: Not disclosed - Chain-transfer agent: Not disclosed - Chain-transfer dosage: Not disclosed - Polymerization temperature: 120 - Pressure: 40 - pH: Not disclosed - Reaction time: 120 - Conversion: Not disclosed - Coagulation conditions: Not disclosed - Post-treatment: Not disclosed - Raw polymer properties: Not disclosed

Technical relevance: This patent is relevant to Low Acrylonitrile NBR polymerization as it provides a composition and method for preparing acrylonitrile butadiene rubber, which includes Low Acrylonitrile NBR. The patent discloses a polymerization temperature of 120 and a pressure of 40.

EP2463335

Patent Details: - Patent Number: EP2463335 - Patent Title: Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ... - Assignee: Not disclosed - Jurisdiction: EP -

Publication Year: 2015-12-02 - Patent Status: Not disclosed - Polymer Type: Nitrile rubber -
Relevance to Low Acrylonitrile NBR: Relevant to the production of nitrile rubber, which includes
Low Acrylonitrile NBR

Polymerization / Synthesis Method: - Polymerization process: Not disclosed - Monomer system:
Not disclosed - Acrylonitrile content: 45 - Butadiene content: Not disclosed - Exact monomer ratio:
Not disclosed - Water amount: 120 - Emulsifier: 20 - Emulsifier loading: Not disclosed - Initiator: 80
- Initiator loading: Not disclosed - Catalyst: Not disclosed - Chain-transfer agent: Not disclosed -
Chain-transfer dosage: Not disclosed - Polymerization temperature: 20 - Pressure: 20 - pH: Not
disclosed - Reaction time: 1 - Conversion: 98 - Coagulation conditions: Not disclosed -
Post-treatment: Not disclosed - Raw polymer properties: Not disclosed

Technical relevance: This patent is relevant to Low Acrylonitrile NBR polymerization as it
provides a composition and method for preparing nitrile rubber, which includes Low Acrylonitrile
NBR. The patent discloses an acrylonitrile content of 45, which is within the range of Low
Acrylonitrile NBR.

US10266615

Patent Details: - Patent Number: US10266615 - Patent Title: Method for preparing
acrylonitrile-butadiene-styrene graft copolymer and ... - Assignee: Not disclosed - Jurisdiction: US
- Publication Year: 2019-04-23 - Patent Status: Not disclosed - Polymer Type:
Acrylonitrile-butadiene-styrene graft copolymer - Relevance to Low Acrylonitrile NBR: Relevant to
the production of acrylonitrile-butadiene-styrene graft copolymer, which includes Low Acrylonitrile
NBR

Polymerization / Synthesis Method: - Polymerization process: Not disclosed - Monomer system:
Not disclosed - Acrylonitrile content: Not disclosed - Butadiene content: Not disclosed - Exact
monomer ratio: Not disclosed - Water amount: Not disclosed - Emulsifier: Not disclosed -
Emulsifier loading: Not disclosed - Initiator: Not disclosed - Initiator loading: Not disclosed -
Catalyst: Not disclosed - Chain-transfer agent: Not disclosed - Chain-transfer dosage: Not
disclosed - Polymerization temperature: Not disclosed - Pressure: Not disclosed - pH: Not
disclosed - Reaction time: Not disclosed - Conversion: 95 - Coagulation conditions: Not disclosed -
Post-treatment: Not disclosed - Raw polymer properties: Not disclosed

Technical relevance: This patent is relevant to Low Acrylonitrile NBR polymerization as it
provides a method for preparing acrylonitrile-butadiene-styrene graft copolymer, which includes
Low Acrylonitrile NBR. The patent discloses a conversion of 95.

EP2949699

Patent Details: - Patent Number: EP2949699 - Patent Title: Process to prepare thermoplastic
molding compositions based on acrylonitrile, ... - Assignee: Not disclosed - Jurisdiction: EP -
Publication Year: 2015-12-02 - Patent Status: Not disclosed - Polymer Type: Thermoplastic
molding composition - Relevance to Low Acrylonitrile NBR: Relevant to the production of

thermoplastic molding compositions, which includes Low Acrylonitrile NBR

Polymerization / Synthesis Method: - Polymerization process: Not disclosed - Monomer system: Not disclosed - Acrylonitrile content: 101 - Butadiene content: 100 - Exact monomer ratio: Not disclosed - Water amount: 200 - Emulsifier: 10 - Emulsifier loading: Not disclosed - Initiator: 5265 - Initiator loading: Not disclosed - Catalyst: Not disclosed - Chain-transfer agent: Not disclosed - Chain-transfer dosage: Not disclosed - Polymerization temperature: 4.96 - Pressure: Not disclosed - pH: Not disclosed - Reaction time: 65 - Conversion: 90 - Coagulation conditions: Not disclosed - Post-treatment: Not disclosed - Raw polymer properties: Not disclosed

Technical relevance: This patent is relevant to Low Acrylonitrile NBR polymerization as it provides a process for preparing thermoplastic molding compositions, which includes Low Acrylonitrile NBR. The patent discloses an acrylonitrile content of 101 and a butadiene content of 100.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

The selected patents disclose various polymerization processes, monomer systems, and acrylonitrile contents. The acrylonitrile content ranges from 34 to 101, with a butadiene content of 100 disclosed in one patent. The polymerization temperatures range from 20 to 120, with a pressure of 20 and 40 disclosed in two patents. The reaction times range from 1 to 120, with a conversion of 90 to 98 disclosed in two patents.

The emulsifier systems and initiator loadings are not consistently disclosed across the patents. However, one patent discloses an emulsifier loading of 20, and another patent discloses an initiator loading of 5265.

The chain-transfer agents and catalysts are not disclosed in any of the patents. However, one patent discloses a chain-transfer dosage of 0.05.

The coagulation conditions and post-treatment methods are not consistently disclosed across the patents. However, one patent discloses a coagulation condition of 9, and another patent discloses a post-treatment method of freeze-crushing.

The raw polymer properties are not consistently disclosed across the patents. However, one patent discloses a raw polymer property of 300.

4. REFERENCES

- EP3037446: [Method for preparing nitrile-based rubber](#)
- EP3466997: [Method for preparing nitrile-based rubber](#)
- EP2966122: [Acrylonitrile butadiene rubber composition, acrylonitrile butadiene rubber ...](#)
- EP2463335: [Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ...](#)
- US10266615: [Method for preparing acrylonitrile-butadiene-styrene graft copolymer and ...](#)

- EP2949699: [Process to prepare thermoplastic molding compositions based on acrylonitrile, ...](#)